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Hugging Face offers a vast ecosystem of libraries that work together like building blocks. You only need to install what is necessary for your specific stage of the AI workflow (data, modeling, or optimization).

Here is a breakdown of the primary libraries, when to use them, and how to install them.

1. The "Big Three" (Core Workflow)

These are the libraries you will use 90% of the time.

Library	When to use it?	Installation
transformers	To download, use, and fine-tune pre-trained models for Text, Vision, or Audio.	<code>pip install transformers</code>
datasets	To download and process massive amounts of data efficiently (text, images, or audio).	<code>pip install datasets</code>
tokenizers	Usually comes with transformers. Use it if you need ultra-fast text preprocessing.	<code>pip install tokenizers</code>

2. Specialized Task Libraries

If you are doing something specific, like generating art or working with large-scale hardware, you'll need these.

- **diffusers**: Used specifically for generative AI like **Stable Diffusion**. If you want to create images or videos, this is the one.
 - *Install*: `pip install diffusers`
- **accelerate**: Used when you want to run your code on multiple GPUs or TPU easily without writing complex PyTorch code.
 - *Install*: `pip install accelerate`

- **peft**: Used for "Parameter-Efficient Fine-Tuning" (like **LoRA**). Use this if you want to fine-tune a huge model (like Llama 3) on a consumer laptop or a single GPU.
 - *Install*: pip install peft

3. Efficiency and Deployment

- **optimum**: Used to speed up your model's performance on specific hardware (Intel, AMD, Nvidia) or for "Quantization" (making models smaller).
 - **evaluate**: A cleaner way to calculate metrics like Accuracy, F1-score, or BLEU during training.
 - *Install*: pip install optimum evaluate
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When to install them (Workflow Guide)

1. Phase 1: Exploration & Inference

- **Goal**: You just want to see if a model works.
- **Install**: transformers
- **Command**: from transformers import pipeline is all you need.

2. Phase 2: Preparing Your Own Data

- **Goal**: You have a custom dataset for your specific project.
- **Install**: datasets
- **Tip**: Use this to load local .csv or .json files into a format that the model understands.

3. Phase 3: Fine-Tuning

- **Goal**: Teaching a model a new skill.
- **Install**: transformers, datasets, accelerate, and peft.
- **Why**: You need the model (transformers), the data (datasets), the training speed (accelerate), and a memory-saving method (peft).

Pro-Tip: The "All-in-One" Installation

If you are starting a new project and want to avoid stopping later to install missing pieces, run this:

Bash

```
pip install transformers datasets accelerate peft diffusers evaluate
```

Would you like me to provide a 5-line code snippet showing how to combine transformers and datasets to run your first model?